# Jongmin Choi

*M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST*

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## RESEARCH INTERESTS

Biological intelligence is the clearest existence proof we have of a general and efficient learner. I believe that networks shaped by its principles rather than by scale alone can become more capable and efficient. My master's research followed this belief in two directions. First, I studied the gap between audio and text and how multimodal models can narrow it while preserving the structure unique to each modality. Second, I asked what the basic unit of computation should be. A biological neuron is not simply a perceptron. It spikes, resets, and is shaped by its membrane channels. I developed spiking neuron algorithms that place these mechanisms within the neuron itself. I am now studying connectors between audio encoders and language-model decoders to reduce information bottlenecks. In parallel, I investigate the mathematical foundations of optimization algorithms for training deep networks.

## EDUCATION

**Korea Advanced Institute of Science and Technology (KAIST)** — *Daejeon, South Korea* *Mar 2025 – Present*

M.S. in Electrical Engineering

Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA **4.06 / 4.30**

**Korea Advanced Institute of Science and Technology (KAIST)** — *Daejeon, South Korea* *Mar 2018 – Feb 2025*

B.S. in Electrical Engineering and Mathematical Sciences (Double Major)

GPA **3.65 / 4.30**

## PUBLICATIONS

*[C] Conference · [P] Preprint · * Equal contribution*

[P] **FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping**
**Jongmin Choi**, Joon Son Chung
Submitted to *Conference on Neural Information Processing Systems (NeurIPS)* · 2026 UNDER REVIEW

[C] **LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy–Schwarz Divergence**
Hyeongkeun Lee*, **Jongmin Choi***, KiHyun Nam, Joon Son Chung
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* · 2026 ORAL

[C] **Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio–Text Modality Gap**
KiHyun Nam*, **Jongmin Choi***, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* · 2026 POSTER

[P] **Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models**
Chaeyoung Jung*, Youngjoon Jang*, **Jongmin Choi**, Joon Son Chung
*arXiv preprint* · 2025

## TEACHING EXPERIENCE

**EE.70038 — Speech Recognition Systems** — *KAIST* *Spring 2026*

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

**EE.40034 — Deep Learning for Visual Understanding** — *KAIST* *Fall 2025*

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

## SKILLS

**Research & ML:** Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (professional working proficiency)

*Last updated: August 2026*